KAMALESAN C

Senior Automotive Test Engineer | HIL/SIL Testing, ECU Validation & Python Automation

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PROFESSIONAL SUMMARY

Accomplished Senior Automotive Test and Validation Engineer with 7.4 years of specialized engineering experience in ECU validation, Hardware-in-the-Loop (HIL) and Software-in-the-Loop (SIL) test automation, and automotive embedded systems. Extensive domain expertise delivering powertrain validation for Mercedes-Benz projects with Capgemini and Bosch Global Software Technologies. Proficient in TraceTronic ECU-TEST, dSPACE SCALEXIO, ETAS LABCAR, CANoe, CANalyzer, INCA, UDS diagnostic protocols (ISO 14229), and Python test script development. Expert in requirement-based verification, regression automation, and defect root-cause analysis.

TECHNICAL EXPERTISE

Testing Disciplines:	HIL Testing, SIL Testing, ECU Validation, System V&V, Regression Testing, Requirements-Based Testing
Test Tools & Hardware:	TraceTronic ECU-TEST, dSPACE SCALEXIO, ETAS LABCAR, SystemDesk, ControlDesk, UDE Debugger
Automotive Protocols:	CAN, LIN, FlexRay, Automotive Ethernet, UDS (ISO 14229), OBD-II, OBDonUDS
Measurement & Calibration:	Vector CANoe, CANalyzer, ETAS INCA, MDA, CANape
Languages & Scripting:	Python (Automated Test Frameworks, PyTest), Embedded C, CAPL, XML, Batch Scripting

PROFESSIONAL EXPERIENCE

Senior Test Engineer — Automotive ECU & HIL/SIL Validation **2019 – Present**
Capgemini Technology Services (Client: Bosch Global Software Technologies) Coimbatore, India

- Spearheaded powertrain ECU validation and HIL test execution for premium OEM programs (Mercedes-Benz).
- Developed and maintained automated test suites in TraceTronic ECU-TEST integrated with dSPACE SCALEXIO and ETAS LABCAR benches.
- Executed comprehensive UDS diagnostic validation, fault injection, and fail-safe recovery testing across CAN/LIN bus networks.
- Authored automated Python tools to parse MDF/A2L measurement files, extract diagnostic DTCs, and auto-generate compliance reports.
- Participated in cross-functional defect triage with ECU software developers, accelerating defect closure and release readiness.

Embedded Test & Software Engineer **2018 – 2019**
Embedded Automation Labs India

- Developed Embedded C firmware and automated regression test harnesses for microcontroller-based controllers.
- Executed hardware communication protocol tests (UART, SPI, I2C) and sensor signal validation using oscilloscopes and logic analyzers.

KEY PROJECTS & ACHIEVEMENTS

Mercedes-Benz Powertrain ECU Validation & Diagnostic Automation
TraceTronic ECU-TEST, dSPACE SCALEXIO, CANoe, INCA, UDS Diagnostic Protocols

- Automated 500+ powertrain validation test scenarios, cutting regression execution cycles by 40%.
- Created automated flashing and diagnostic verification scripts verifying ECU software build integrity.

Python-Based SIL Test Automation Framework
Python, Simulation Models, CANalyzer API, Automated Test Reporting

- Constructed Python simulation test runner enabling developers to run virtual ECU validation ahead of physical bench allocation.

EDUCATION & CREDENTIALS

- **Master of Computer Applications (M.C.A)** — Bharathidasan University
- **Automotive Diagnostics (UDS & OBD-II)** Validation Training
- **Python for Automotive Test Automation** Certification

MOBILITY & WORK AUTHORIZATION

- **Mobility:** Open to International Relocation & Worldwide Remote
- **Sponsorship:** Requires work visa for EU/US on-site positions
- **Languages:** English (Professional), Tamil (Native)